

Group Assignment 1: A Bayesian Reanalysis of the French Short Boredom Proneness Scale

Nikola Sekulovski & Maarten Marsman

Information about the assignment

In this group assignment, you will **reanalyze an existing dataset** using the Bayesian the graphical modeling methodology introduced in the workshop. You are provided with a short study description and a **fully specified Methods section**. Your task is to:

- Implement the analysis exactly as implied by the Methods section
- Make justified analytical decisions where choices are left open
- Write a very brief **Results section** where the results are summarized.

You will work with the **French subsample** ($N = 490$) from the original study, using **18 ordinal items** from the **Short Boredom Proneness Scale (SBPS)** and the **Curiosity and Exploration Inventory–II (CEI-II)**.

Introduction

Boredom proneness has been conceptualized as a trait-like tendency to experience boredom frequently and intensely, and has been linked to a broad range of psychological outcomes. Boredom proneness has been conceptualized as a trait-like tendency to experience boredom frequently and intensely, and has been linked to a broad range of psychological outcomes. The Short Boredom Proneness Scale (SBPS) was developed as a concise unidimensional measure of boredom proneness and has since been translated into multiple languages.

Recent work has increasingly adopted a network perspective, in which psychological constructs are modeled as systems of interacting components rather than as manifestations of latent variables. Within this framework, network models offer a flexible way to examine conditional associations between individual questionnaire items while controlling for all other variables in the system.

Therefore, in the original validation of the French version of the SBPS, Martarelli et al. (2022) examined the relationships between boredom proneness, curiosity, and exploration using a regularized frequentist network approach. Their analysis suggested positive associations between boredom and curiosity items and negative associations between boredom and exploration items.

The aim of the present study is to reanalyze the French subsample of the original dataset using Bayesian graphical modeling. By adopting a Bayesian approach, we explicitly quantify uncertainty in network structure and parameters, and we distinguish between absence of evidence and evidence of absence for network edges. In particular, we estimate a Bayesian ordinal network model that directly takes into account the ordinal measurement level of the variables.

Method

Data

The dataset consists of responses from $N = 490$ French-speaking participants ($M_{age} = 33.8$; 73% females and 27% males). The network includes 18 items, all measured on a 7-point Likert scale:

- 8 items from the Short Boredom Proneness Scale (SBPS)
- 10 items from the Curiosity and Exploration Inventory–II (CEI-II)

Missing data was handled by listwise deletion.

Model Specification

To model conditional associations between items, we estimated a Bayesian Ordinal Markov Random Field (OMRF, Marsman et al., 2025). This model represents each item as a node in an undirected network, with edges encoding conditional dependence relationships between item pairs while controlling for all remaining variables in the network.

Bayesian model averaging was employed to integrate over uncertainty in network structure. As a result, inference was not based on a single selected network, but on the posterior distribution over possible network structures.

Prior Distributions

Structure Priors

The network structure was assigned a Bernoulli prior on edge inclusion indicators. In the primary analysis, default prior settings were used. To assess robustness, a prior sensitivity analysis was conducted by repeating the analysis under alternative structure priors reflecting different expectations about network density, ranging from relatively sparse to relatively dense networks.

Parameter Priors

Conditional association parameters were assigned default priors that regularize edge weights toward zero. To examine the influence of these assumptions, a sensitivity analysis was performed by varying the scale of the parameter priors, allowing for both more concentrated and more diffuse prior distributions.

Inference and Evidence Quantification

Evidence for the presence or absence of edges was quantified using inclusion Bayes factors, comparing models in which a given edge was included to models in which it was excluded. Bayes factors were used to classify edges as showing evidence for inclusion, evidence for exclusion, or inconclusive evidence. We use the edge evidence plots to visualize which edges have evidence for inclusion, exclusion, or remain inconclusive. In this paper, we use an inclusion Bayes factor of 10 as the threshold for strong evidence for inclusion, and a Bayes factor of 0.1 as the threshold for strong evidence for exclusion, the values in between are classified as inconclusive evidence.

Posterior summaries of edge parameters were obtained from the model-averaged posterior distribution. Centrality measures were not estimated or interpreted.

Software

We used the `easybgm` R package (Huth et al., 2024) to estimate the Bayesian OMRF model and conduct the analyses.

Results

Network Estimation

This section should report general characteristics of the estimated network, including the number of edges for which there was conclusive evidence, the number of possible edges, and an overall description of the estimated network structure.

Edge Evidence

This section should describe the results of the Bayes factor analysis at the edge level. Report the edge evidence plot. Particular attention should be paid to patterns of associations within and between the SBPS and CEI-II items. Edge evidence plots or summary tables should be referenced here.

Edge Parameter Estimates and Uncertainty

This section should summarize the estimated edge parameters for edges with non-negligible posterior support. Describe the direction and relative magnitude of the associations, focusing on relationships between boredom, curiosity, and exploration items. Interpretations should remain descriptive and avoid causal language. Also discuss uncertainty in the estimated parameters.

Prior Sensitivity Analysis (if time permits)

This section should compare the primary analysis to the sensitivity analyses. Discuss whether the global conclusions about edge inclusion, exclusion, and parameter estimates are stable across different structure and parameter priors, and highlight any results that are sensitive to prior assumptions.

References

- Huth, K., Keetelaar, S., Sekulovski, N., van den Bergh, D., & Marsman, M. (2024). Simplifying Bayesian analysis of graphical models for the social sciences with easybgm: A user-friendly R-package. *advances.in/psychology*, *e66366* doi: 10.56296/aip00010
- Martarelli, C. S., Baillifard, A., & Audrin, C. (2022). A trait-based network perspective on the validation of the french short boredom proneness scale. *European Journal of Psychological Assessment*. doi: 10.1027/1015-5759/a000718346
- Marsman, M., van den Bergh, D., & Haslbeck, J. M. B. (2025). Bayesian analysis of the ordinal Markov random field. *Psychometrika*, *90* , 146–182. doi: 10.1017/psy.2024.4